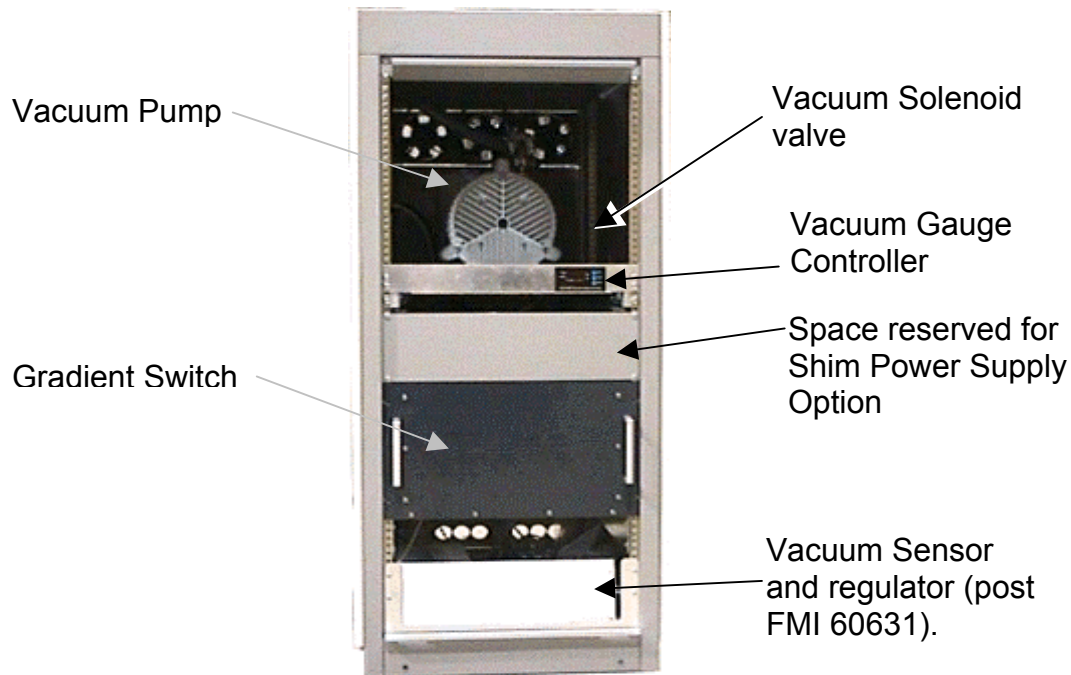


TABLE OF CONTENTS

Table of Contents	1
1- Time and Tools and Equipment	2
2- Vacuum Subsystem Overview	2
3- Vacuum Gage Calibration/program check.....	3
4- Vacuum Pump Down Check.....	5
5- Verify Pressure Regulation Check (150 to 190 Torr)	6
Revision History	7

Description – This procedure checks the operation of the vacuum pump and vacuum regulator in the Twin Accessory Cabinet, the vacuum sensor in the Rear Pedestal, and the vacuum fittings and seals that make up the vacuum enclosure. Estimation of time required to perform this procedure is 0.5 hours.



TWIN ACCESSORY CABINET (TAC)
ILLUSTRATION 1-1

1- TIME AND TOOLS AND EQUIPMENT

Tools

Timing Device (watch or clock with second hand)

2- VACUUM SUBSYSTEM OVERVIEW

Note

There are two styles of vacuum sensor's and gauge controllers. The old style gauge controller and sensor requires calibration, the new style is pre-calibrated and does not require/allow field adjustments. See Illustration below for identifying sensor type.



The vacuum pump is a dry scroll pump. This pump provides the suction to pull down the pressure in the enclosure. A vacuum sensor sends vacuum pressure readings to the vacuum gauge controller in the Twin Accessory Cabinet. This controller interprets the signal and displays the vacuum pressure reading. In addition, the controller has 2 programmable setpoints with low and high values. Set point 1 High 190 Torr and Low 150 Torr sends a signal to the vacuum regulator box. This intern routes power to the solenoid valve, closing it when it hits 150 Torr and opening it at 190 Torr. Set Point 2 High 200 Torr and Low 130 Torr sends a signal to the vacuum regulator box. This intern routes power to the vacuum pump, turning it off if the pressure hits 130 Torr and back on when it hits 200 Torr. This would only occur if the solenoid valve failed to control the pressure. Finally, the vacuum pressure is sent to the magnet monitor from the vacuum gauge controller.

WARNING

The Vacuum pump should never be operated to a value of less than 100 Torr when power is applied to the RF coil. Serious damage to the RF coil could result.

3- USER BUTTON DESCRIPTIONS

- ◆ **Select Button** – Advances Led Bar to appropriate controller function.
- ◆ **Gauge Select Button** – Toggles display between Gauge 1 & Gauge 2 inputs. Note: This system will use only Gauge 1.
- ◆ **Raise/Lower Buttons** – Used to raise or lower setpoint limits, atmosphere setting and zero calibration adjustment. The new style Vacuum Gauge Controller disables the functionality of this button. Note: We will not use zero calibration adjustment for this system.

4- VACUUM GAGE CALIBRATION/PROGRAM CHECK

Note

The following adjustment procedure will only work on the old style vacuum controller, model **926**. This procedure can be used to verify proper programming of the new vacuum gauge. The *Raise/Lower* button is disabled on the new vacuum gauge, no changes can be made.

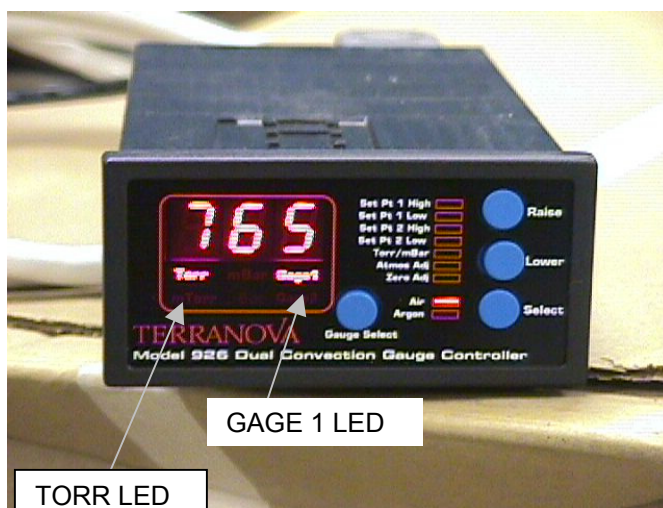
1. Verify vacuum pump is off. Make sure vacuum system is at atmospheric pressure by opening the flange seal at the pump and allowing air to enter system for 20 seconds.
2. Refer to Table 4-1 for atmospheric pressure setting value. Use the systems altitude to select atmospheric pressure in units of Torr.

Altitude Above Sea Level		Barometer Pressure	Altitude Above Sea Level		Barometric Pressure
Feet	Meters	Torr	Feet	Meters	Torr
-1000	-305	790	4000	1220	655
-500	-153	775	4500	1373	645
0	0	760	5000	1526	630
500	153	745	6000	1831	610
1000	305	735	7000	2136	585
1500	458	720	8000	2441	565
2000	610	705	9000	2746	545
2500	763	695	10000	3050	520
3000	915	680	15000	4577	430
3500	1068	670	20000	6102	350

ATMOSPHERIC PRESSURE VS ALTITUDE TABLE

Table 4-1

3. Go to Twin Accessory Cabinet and verify the Vacuum Gauge Controller (VGC) is powered. Verify the VGC display shows a pressure, and the **Torr** and **Gage1** LED's are lit. See Illustration 4-1. If they are not lit, press **SELECT** button to advance led bar and stop at "Torr/mBar" location. Press **Raise** or Lower to change to "Torr". To change to Gage 1, press **Gauge Select** button.



VACUUM GAUGE CONTROLLER
ILLUSTRATION 4-1

4. Use Table 4-2 and make sure controller is programmed correctly. Use Raise/Lower buttons (Model 926 only) to program values.

Setting	Value	Units
Set Pt 1 High	190	Torr
Set Pt 1 Low	150	Torr
Set Pt 2 High	200	Torr
Set Pt 2 Low	130	Torr
Torr/mBar	Torr	N/A
Atmos Adj	Value from Step 2	Torr
Zero Adj	N/A	N/A
Air/Argon	Air	N/A

VACUUM GAUGE SETTINGS

TABLE 4-2

5. Verify that the "Select" button is switched to "AIR" to properly monitor the vacuum.

5- VACUUM PUMP DOWN CHECK

1. Use a clock or a watch that displays seconds.
2. Start timing after turning on Vacuum Pump. The power switch is located on the side of vacuum pump inside of Twin Accessory Cabinet.
3. Monitor Vacuum Gauge Controller reading; the pressure should reduce to 150 Torr within 5 minutes. If 150 Torr is not reached, check for vacuum leaks and secure hose connections. Refer to vacuum troubleshooting procedure.

6- VERIFY PRESSURE REGULATION CHECK (150 TO 190 TORR)

1. After reaching 150 Torr, the solenoid valve should close and the pressure should start rising.
2. When 190 Torr is reached, the valve should open and pressure should start decreasing. Nominal recovery time for vacuum to decay from 150 Torr to 190 Torr is around 60 seconds. If the decay time is around 30 seconds or less check for vacuum leaks.
3. Watch this for at least 3 cycles for proper operation.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
A	Aug. 13, 2001	D. Davis/P. Senski	Initial release for TwinSpeed system.
B	Aug. 23, 2001	P. Senski/ J. Wolak	Refined procedure
0	Nov 19	P. Kargard	Added warning about possible RF coil failure
1	Nov 20	P. Kargard	Fixed discrepancy error for vacuum setting
2	Feb 15, 2003	P. Kargard	Information for new style sensor and controller